

The Limits of Your World Model Mean the Limits of Your Language Model: Embodied Reasoning through a Predictive Action Boundary

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Abstract

Large language models exhibit strong semantic and logical reasoning, yet they remain limited in judging which actions are likely under the current visual situation and action history. World models, by contrast, predict future actions from observed experience, but how to connect these predictions to high-level action judgments consistent with long-horizon task context remains underexplored. We introduce **EGO (Embodiment Grounded Ontology)**, a world-model-based framework in which a VLM selects a task-appropriate action within a boundary predicted from the current observation and action history by a separately trained V-JEPA2-based action anticipation model. Rather than generating actions directly from a broad linguistic action space, the VLM aligns high-level reasoning with action possibilities supported by visual experience. On real-world egocentric interactive video, diagnostics and interventions show that EGO increases reliance on action history and strengthens belief-conditioned action selection beyond action supervision alone. The resulting model also improves multi-step video procedure anticipation. By explicitly aligning visual prediction and symbolic reasoning rather than fusing them end to end, EGO offers a new approach to embodied reasoning.

Introduction

Language models demonstrate strong semantic and logical reasoning, further extended by reinforcement-learning-based post-training from mathematics to physical commonsense and action judgment (Wei et al. 2022; Eldan and Li 2023; Ouyang et al. 2022; Guo et al. 2025; Azzolini et al. 2025). Yet they cannot reliably determine whether a contextually relevant action is likely under the current visual situation and action history. **Embodied reasoning**, selecting actions supported by visual experience, therefore remains an open problem.

World models operationalize the human capacity to predict physical changes from experience (Ha and Schmidhuber 2018), but learned dynamics mainly support short-term state prediction and low-level control (Hafner et al. 2025; Hansen, Su, and Wang 2024; Zhou et al. 2024). High-level, long-horizon action judgments for embodied agents in complex environments remain underexplored. Figure 1 illustrates these complementary limitations.

Existing integrations do not fully close this gap. **LLM as World Model** methods generate and evaluate futures using

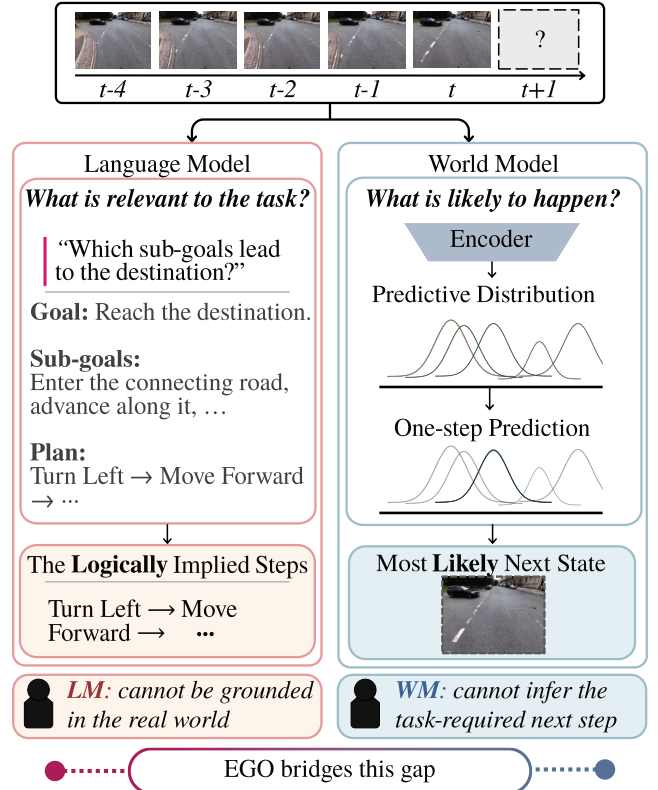


Figure 1: **Problem formulation.** Language models infer task-relevant actions but may overlook visual constraints, whereas world models predict visually likely actions without long-horizon task context. EGO bridges this gap.

the same model’s prior, without visual evidence (Hao et al. 2023; Wang et al. 2025). **Language as World Representation** methods abstract states into language (Lin et al. 2024; Chen et al. 2025), losing visual detail while accumulating cost. We ask: can a language model reason within a boundary predicted by a world model?

We propose **EGO (Embodiment Grounded Ontology)**, which progressively aligns independently trained world and language models. EGO is motivated by the brain’s complementary functional specialization: spatial and nonverbal pro-

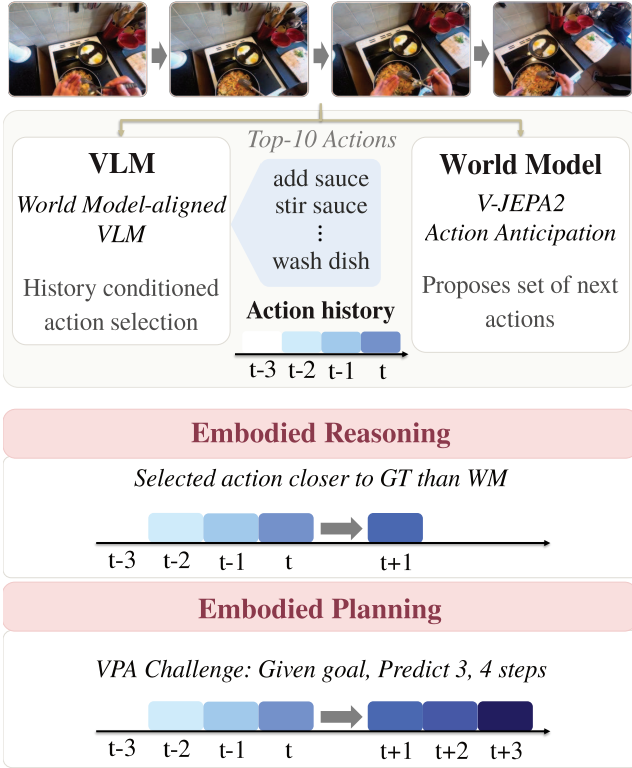


Figure 2: **EGO overview**. A frozen video world model forms an action boundary from egocentric observations and history; an aligned VLM selects a task-consistent action within it. Evaluation covers single-step embodied reasoning and multi-step procedure anticipation.

cessing is relatively right-lateralized, whereas language and prior knowledge are relatively left-lateralized (Cai, Van der Haegen, and Brysbaert 2013). The cognitive intuition that representations formed from new sensory experience precede their interpretation and use in a linguistic system motivates EGO’s modular two-stage alignment: action predictions are first formed from nonverbal visual experience and then passed in one direction as an alignment signal for high-level language reasoning (Goldberg and Costa 1981).

Concretely, a V-JEPA2-based action anticipation model, pretrained on large-scale video, predicts the Top- K next actions from the current observation and action history (Assran et al. 2025). These candidates form a *predictive action boundary* within which the VLM selects a task-consistent action. EGO aligns the VLM with this prior through **Prospection** and **Retrospection**.

Prospection uses candidate-conditional likelihoods to train the VLM to discriminate the task-consistent action within the boundary. Because action supervision alone does not align the supporting reasoning, **Retrospection** reconstructs the task judgment from subsequent actions and projects it into reasoning and a task belief grounded only in decision-time evidence. EGO thereby learns both which action to select and why, without future information at inference time (Figure 2).

We test how the two stages change the model’s reliance on action history, observable evidence, and task belief, and whether these changes accompany improved action selection. Against a control trained on the same correct actions without the boundary, EGO isolates the grounding effect of the world-model prior from mere target supervision. We further examine generated reasoning traces and transfer to multi-step video procedure anticipation.

Our contributions are:

- We introduce **EGO**, a modular framework that grounds VLM action selection in a boundary predicted by an independently trained V-JEPA2-based action anticipation model without generating future frames.
- We propose **Prospection–Retrospection**: Prospection learns candidate discrimination, while Retrospection projects hindsight-confirmed judgments to align reasoning, task belief, and action.
- We show on real-world egocentric interactive video, through quantitative diagnostics and generated reasoning traces, that EGO grounds judgment in visual predictions to support task-consistent action selection and transfer across multiple steps.

Related Work

World Models for Action Prediction

World models select actions by predicting action-conditioned future states (Ha and Schmidhuber 2018), but their successes have largely centered on low-level control in interactive simulations, limiting their extension to real-world activities (Hafner et al. 2025). Recent work instead learns environmental changes and action outcomes from large-scale real-world observations.

Generative video world models produce future scenes or long-horizon rollouts (Yang et al. 2024; Brooks et al. 2024; Agarwal et al. 2025), but modeling task-irrelevant visual details increases prediction cost and error. The JEPA family instead predicts abstract components in representation space (LeCun 2022). V-JEPA captures temporal and motion structure in video (Bardet et al. 2024), while V-JEPA2 scales this approach for motion understanding, action anticipation, and physical reasoning (Assran et al. 2025). However, its next-action distribution is not a verifier of physical feasibility, but a predictive prior over behaviorally plausible actions. It can narrow the action space without determining which action fits the task context.

Post-Training Language Models for Reasoning

Post-training aligns pretrained reasoning with desired behavior, from InstructGPT’s human-preference reward modeling (Ouyang et al. 2022) to DPO and KTO’s direct preference optimization (Rafailov et al. 2023; Ethayarajh et al. 2024). Reliance on human judgments creates labeling and scalability challenges in embodied domains, where the appropriate decision criterion varies across states.

Verifiable rewards automate supervision but require pre-defined criteria. DeepSeek-R1 verifies explicit answers in domains such as mathematics and code (Guo et al. 2025),

while Cosmos-Reason1 extends structured automatic scoring to physical commonsense and embodied reasoning. Neither aligns language reasoning with state-dependent action possibilities learned from visual experience.

World–Language Model Integration

World–language model integration differs in where future predictions are formed. **LLM as World Model** methods generate and evaluate action-conditioned futures in language space without a separate visual predictor (Hao et al. 2023; Wang et al. 2025), whereas **Language as World Representation** methods encode states and action-induced changes as language sequences (Lin et al. 2024). Both therefore depend on language-model priors and incur accumulating cost and error through repeated generation.

VLWM learns interleaved action–state trajectories from natural videos for reactive selection and rollout planning, but forms both world predictions and actions within the same language-based model (Chen et al. 2025).

Other approaches generate visual futures for planning or VLM post-training (Feng et al. 2025; Jia et al. 2026). World2VLM distills motion-conditioned futures into dynamic spatial reasoning, removing inference-time generation, but does not directly address task-conditioned action selection (Zhang et al. 2026). These methods still generate futures before evaluation or distillation. An alternative is to connect predicted action candidates directly to the VLM’s selection problem without mediating through future scenes.

ADV uses a diffusion action expert to propose continuous action chunks that the VLM reranks at inference time (Zhao et al. 2026). In contrast, EGO uses an independently trained representation-based video predictor to construct a discrete Top- K predictive action boundary from the current observation and completed-action history, without generating future scenes. The visually learned action prior serves as both a VLM post-training signal and an inference-time selection boundary rather than only for test-time reranking.

Methods

Overview

EGO first grounds the VLM’s action space in candidates from a world-model prior with a frozen V-JEPA2 backbone, then aligns the VLM through **Prospection** and **Retrospection**. We use *world-model prior* in the limited predictive sense of a model learned from visual experience that estimates behaviorally plausible future actions; EGO does not assume explicit state-transition simulation or counterfactual rollouts. World-model grounding constructs a likely-action boundary within which Prospection learns task-conditioned candidate discrimination. Retrospection aligns reasoning, task belief, and action through hindsight-confirmed decision-time projection, while replay preserves discrimination. EGO thereby learns to select and justify an action within the predicted boundary. We instantiate EGO with a V-JEPA2-based action anticipation model; generalization to other predictive models remains future work. Figure 3 summarizes the pipeline.

World-Model Grounding

EGO grounds the VLM’s choice set in a behavioral prior from a separately trained video action anticipation model. At the same hierarchy level, the model predicts the first action after the observed reference action ends. Unlike V-JEPA2’s public downstream classifier, it receives no target-action frames. An ordered history of label-free visual features from completed segments captures procedural progress beyond the current observation.

Let $x_{\leq t}$ denote observations available by decision time t , $Z_{< t}$ the visual history of completed action segments, and a a possible next action. The world-model prior and predictive action boundary are

$$q_{\text{WM}}(a \mid x_{\leq t}, Z_{< t}), \quad D_t = \text{TopK}_K[q_{\text{WM}}(\cdot \mid x_{\leq t}, Z_{< t})]. \quad (1)$$

Here, q_{WM} is the next-action distribution and D_t contains its K highest-scoring actions. *Candidate coverage*, the rate at which the annotated next action lies in D_t , upper-bounds EGO’s end-to-end accuracy because the VLM selects only within this boundary.

Prospection

Prospection trains the VLM to discriminate the annotated next action within D_t . World-model scores and ranks are hidden, and candidates are shuffled so that selection must rely on the current observation and completed-action history. Define the decision-time context as $c_t = (x_{\leq t}, H_{< t}, D_t)$, where $H_{< t}$ records completed actions. The VLM π_θ generates reasoning r_t , task belief b_t , and an action $a_t \in D_t$:

$$(r_t, b_t, a_t) \sim \pi_\theta(\cdot \mid c_t). \quad (2)$$

For optimization, each candidate is appended as an action completion to the same context, and only its action span is scored. For $a = (a_1, \dots, a_{|a|})$, Prospection uses the mean action-token log-likelihood and candidate cross-entropy:

$$s_\theta(a; c_t) = \frac{1}{|a|} \sum_{j=1}^{|a|} \log \pi_\theta(a_j \mid c_t, a_{< j}), \quad (3)$$

$$\mathcal{L}_{\text{Pro}} = -\log \frac{\exp s_\theta(a_t^*; c_t)}{\sum_{\tilde{a} \in D_t} \exp s_\theta(\tilde{a}; c_t)}.$$

Here, a_j and $a_{< j}$ are the j th action token and its preceding tokens, $|a|$ is the candidate length, a_t^* is the observed next action, and $\tilde{a}$ ranges over D_t . Mean scoring reduces length bias. Training uses only the annotated action within the boundary and yields θ_{Pro} . Prospection thus learns candidate reranking rather than copying WM Top-1, but action-only supervision does not ensure that reasoning and task belief support selection, motivating Retrospection.

Retrospection

Retrospection interprets the task meaning revealed after a decision and reconstructs the reasoning and task belief that should support the observed action using only decision-time evidence. Subsequent events construct supervision; they are not inputs for relearning candidate discrimination.

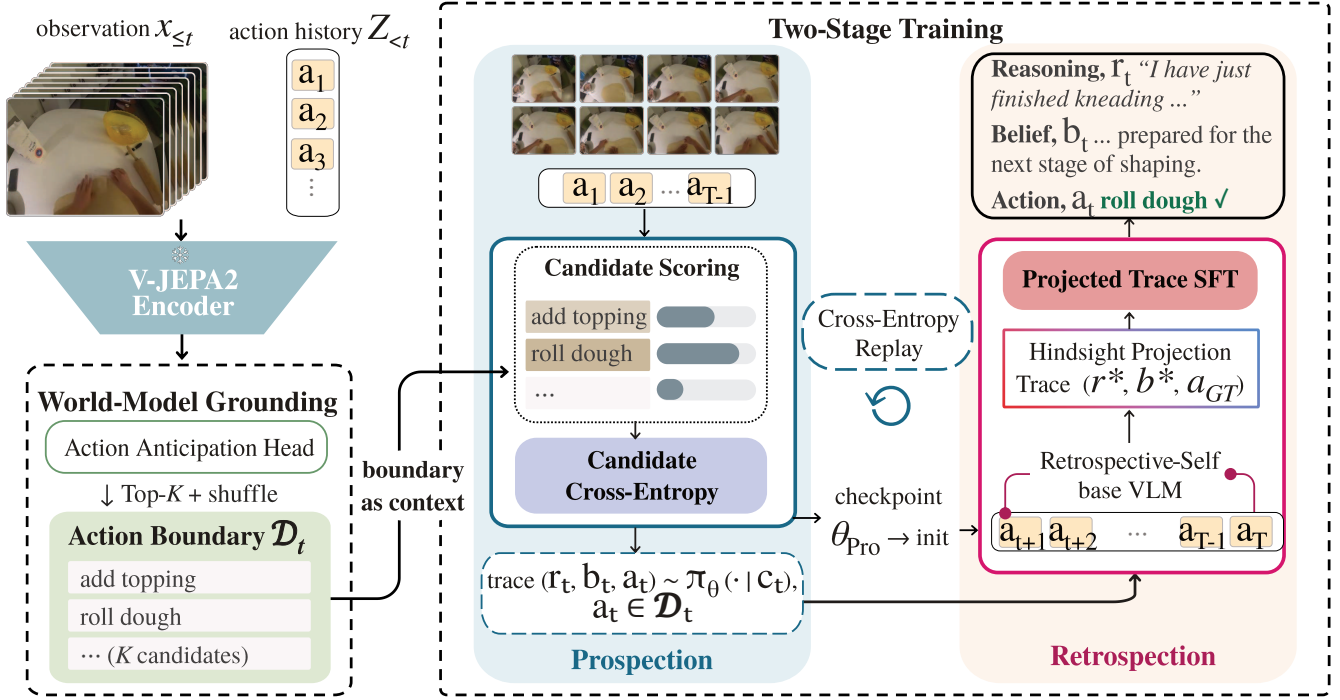


Figure 3: **EGO training pipeline.** World-model grounding constructs a shuffled Top- K action boundary from egocentric observations and completed-action history. Prospection learns candidate discrimination, while Retrospection aligns reasoning, task belief, and action through projected traces. Replay preserves Prospection’s discrimination criterion, and subsequent events are used only for training.

Hindsight projection. A parameter-frozen copy of the base VLM, the *retrospective self* $\mathcal{S}_R$, performs hindsight interpretation and decision-time projection offline. It first interprets the limited annotated action sequence $\tau_{>t}$ after the decision as a hindsight belief b_t^H , then projects this belief with c_t and a_t^* into a supervision trace:

$$\begin{aligned} b_t^H &= \mathcal{S}_R^{\text{reflect}}(\tau_{>t}), \\ \hat{y}_t &= \mathcal{S}_R^{\text{project}}(c_t, b_t^H, a_t^*) = (\hat{r}_t, \hat{b}_t, a_t^*). \end{aligned} \quad (4)$$

The two superscripts denote calls to the same retrospective self. $\hat{r}_t$ explains how decision-time evidence supports the action, while $\hat{b}_t$ summarizes the task state inferred from that evidence. Thus, projection converts a hindsight-confirmed judgment into a trace explainable without future information. The actual action a_t^* is used only as an offline target, and a quality gate rejects traces that repeat it without evidence or mention future events.

Accepted traces jointly supervise reasoning, belief, and action. Let $\mathcal{U} = \{r, b, a\}$ and $\mathcal{I}_u$ denote the token positions of field u . The field and Retrospection losses are

$$\begin{aligned} \mathcal{L}_u &= -\frac{1}{|\mathcal{I}_u|} \sum_{j \in \mathcal{I}_u} \log \pi_\theta(\hat{y}_{t,j} | c_t, \hat{y}_{t,<j}), \quad u \in \mathcal{U}, \\ \mathcal{L}_{\text{Retro}} &= w_r \mathcal{L}_r + w_b \mathcal{L}_b + w_a \mathcal{L}_a, \quad w_b > w_r > w_a. \end{aligned} \quad (5)$$

Here, $\hat{y}_{t,j}$ is the target token and $\hat{y}_{t,<j}$ its prefix. Belief receives the greatest weight, reasoning an intermediate weight,

and action the smallest because Prospection already optimizes it directly.

Prospection Replay. Projected-trace SFT may weaken Prospection’s discrimination criterion. During Retrospection, we therefore add a candidate-selection micro-step using Equation (3) after each SFT micro-step with probability ρ . Let R_m be the m th SFT step, P_m its Prospection step, and $z_m \sim \text{Bernoulli}(\rho)$:

$$\Gamma_m = \begin{cases} (R_m, P_m), & z_m = 1, \\ (R_m), & z_m = 0, \end{cases} \quad \Gamma = \Gamma_1 \oplus \dots \oplus \Gamma_M. \quad (6)$$

Here, M is the number of SFT steps and $\oplus$ denotes concatenation. Because every R_m is retained and P_m is only added, replay preserves projected-trace exposure without directly competing objectives. Both use the same optimizer and gradient accumulation. A paired comparison varying only ρ evaluates forgetting.

Decision-Time Reasoning. At inference, $\pi_{\theta_{\text{Retro}}}$ receives only $c_t = (x_{\leq t}, H_{<t}, D_t)$. It generates reasoning and a task belief, then selects an action within D_t conditioned on them. The subsequent trajectory used by Retrospection never expands the policy’s inference-time information. The final VLM thus preserves Prospection’s candidate discrimination while learning to make reasoning and task belief inform action selection.

		COMPLETED HISTORY ... → <i>add rice</i>	WORLD-MODEL CANDIDATE BOUNDARY - shuffled, ranks hidden <i>check heat · stir ingredient</i> ^{WM Top-1} · <i>add rice</i> ^{GT} · ... (10)
Policy	reasoning trace, BELIEF, and selected action		
Base VLM	“I am currently stirring a yellow mixture in a pan on the stove, and I have just transferred some of it to a smaller pot. The mixture appears to be thickening, and I need to ensure it cooks evenly. . . to continue stirring the main pan to prevent sticking.” ! BELIEF Stir the mixture in the pan to ensure even cooking. ! ACTION stir ingredient × [one spatial cue; no hand/tool cue; history not used]		
Prospection	“The current action involves adding rice to the pan, which is part of a repetitive cooking process. The candidate actions include ‘add rice’, which aligns with the ongoing task of adding rice to the pan. The other candidates are unrelated to the current cooking activity.” ! BELIEF Adding rice to the pan as part of the cooking process. ! ACTION add rice ✓ [cites the repeated run in the history; no spatial or hand/tool cue]		
EGO	“I have just finished adding the final portion of rice to the pan and am now holding the ladle over it, with the other hand still near the pot of batter. The rice is visibly settled in the pan, and I am pausing before stirring, likely to let it sit for a moment or to assess the consistency. The batter is still in the pot nearby, and I have not yet begun to mix or incorporate it into the rice. . . ” ! BELIEF I have just added some of the rice and am pausing before beginning to mix it with the batter. ! ACTION add rice ✓ [two hand/tool cues and two spatial cues; the belief states the trajectory position rather than the action]		

Table 1: **First example.** We compare how Base, Prospection, and EGO justify action selection under the same input. Underlining marks spatial cues and boldface marks hand/tool cues. The example was selected using a fixed criterion, and only the key portions of each trace are shown.

Experiments

We evaluate EGO on the first-person evaluation cohort of the Ego4D GoalStep held-out split (Grauman et al. 2022; Song et al. 2023). Our experiments examine whether the predictive action boundary improves action selection, how Prospection and Retrospection change the model’s judgment, and whether the learned capabilities extend across boundary sizes and multi-step anticipation.

Evaluation Protocol

We split videos at the video level to prevent overlap with training data. By default, the video predictive model estimates $K = 10$ next-action candidates from the current observation and completed-action history. The VLM receives 8 observation frames, the completed-action history, and the candidate boundary. Candidates are shuffled, and their world-model scores and ranks are hidden to prevent selection by predictive order.

We compare five conditions. *WM Top-1* selects the leading world-model candidate without a VLM, providing a rank-copying floor. *Base VLM* is the pre-post-training reference. *No-Boundary* is trained on the annotated action without the candidate boundary, whereas *Prospection* learns to discriminate the annotated action within that boundary. *EGO* is the final policy after Prospection and Retrospection. All VLM conditions use the same Qwen3-VL-8B backbone and evaluation prompt. The central comparison between No-Boundary and Prospection holds completed-action history, target exposure, and training budget fixed, isolating the effect of candidate-aligned learning.

Our primary metrics decompose performance into world-model coverage and VLM selection. Coverage@ K measures whether the annotated next action appears in the Top- K

boundary; Within-Boundary Accuracy measures correct selection among covered examples; and End-to-End Accuracy includes boundary misses. Because Ego4D provides a single observed continuation at each decision, these metrics measure agreement with the annotated next action rather than unique feasibility or optimality. Invalid outputs count as incorrect, and video-clustered paired bootstrap yields confidence intervals. Additional intervention and trace metrics are introduced with their corresponding analyses.

First Example

Table 1 contrasts how Base, Prospection, and EGO use evidence under the same observation, completed-action history, and candidate boundary. A criterion fixed before inspection selects 38 cases from 25 videos. The example is illustrative rather than standalone evidence; subsequent aggregate and intervention analyses test the pattern over the full evaluation set.

Two-Stage Learning Dynamics

Figure 4 shows distinct learning dynamics across stages. Prospection improves Within-Boundary Accuracy while No-Boundary remains near its initial level. Because both conditions receive the same history, targets, and training budget, the difference indicates that the boundary provides a task-contextual discrimination signal beyond target exposure alone.

We define *WM-Correct Retention* as the rate at which a model preserves the world model’s Top-1 action when that prediction is correct. Retrospection primarily strengthens this retention and belief–action coupling rather than further improving accuracy. Belief therefore increasingly informs selection, although stronger coupling does not itself imply

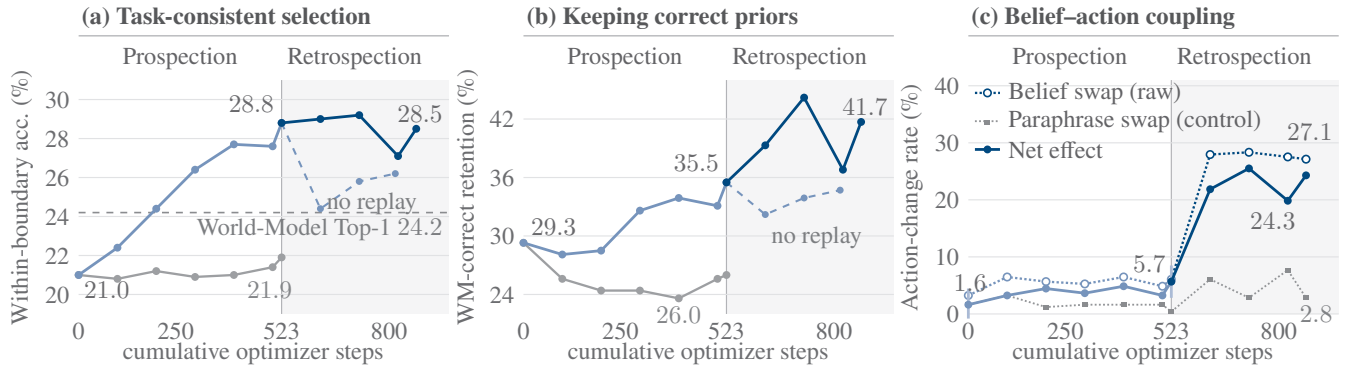


Figure 4: **Two-stage learning dynamics.** Prospection improves action selection within the candidate boundary (a), whereas Retrospection strengthens retention of correct world-model predictions (b) and the influence of task belief on action selection (c). Dotted curves show that Retrospection without replay weakens capabilities learned in the preceding stage.

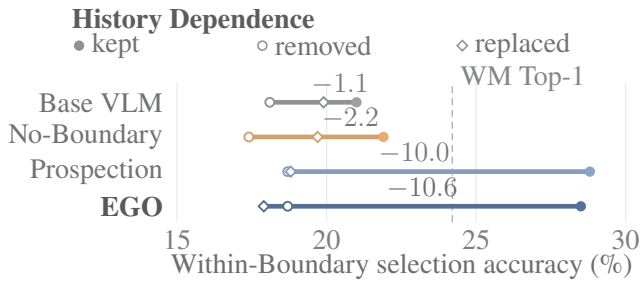


Figure 5: **History dependence.** We hold the observation and candidates fixed and either remove the completed-action history or replace it with history from another video. Accuracy drops substantially only for Prospection and EGO, showing that both models use the action trajectory to discriminate visually similar candidates.

greater correctness. Without replay, both selection accuracy and WM-Correct Retention decline, indicating interference between the objectives. Overall, Prospection learns candidate discrimination, Retrospection couples belief to selection, and replay mitigates forgetting.

Embodied Reasoning

We examine embodied reasoning through predictive-prior retention, functional dependence on action history and task belief, and observable-evidence use in generated reasoning.

Predictive prior and action history. When WM Top-1 is correct, No-Boundary retains it less often than Base, whereas Prospection increases WM-Correct Retention and EGO strengthens it further (Figure 4(b)). To measure *History Dependence*, we hold the observation and candidates fixed and either remove the completed-action history or replace it with history from another video. Replacement reduces accuracy by 10.0 pp for Prospection and 10.6 pp for EGO, but little for Base or No-Boundary (Figure 5). Because visually plausible candidates can be difficult to distinguish from frames alone, Prospection learns to use action history, and Retrospection reinforces this dependence.

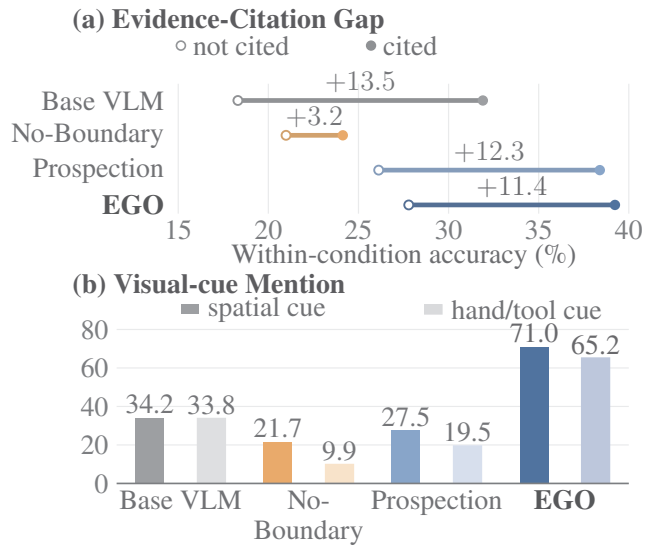


Figure 6: **Evidence citation and visual cues.** (a) compares accuracy between traces that cite a completed action and those that do not. (b) reports the mention rates of spatial cues describing object locations and relations and hand/tool cues referring to hands or tools. Both cue types increase for EGO.

The effect therefore reflects candidate-aligned discrimination rather than merely receiving candidates at inference.

Evidence citation and visual cues. The *Evidence-Citation Gap* measures the accuracy difference between traces that cite a completed action and those that do not. It is positive for Base and both aligned conditions, whereas only No-Boundary’s interval includes 0 (Figure 6(a)). *Visual-Cue Mention* measures the proportion of traces detected as referencing spatial or hand/tool cues. Both rates increase substantially for EGO, consistent with Retrospection’s projected-trace supervision emphasizing decision-time evidence (Figure 6(b)). These trace results remain observational associations.

Method	$T=3$			$T=4$		
	Seq-EM	Pos.Acc	Set IoU	Seq-EM	Pos.Acc	Set IoU
WM Top-1	0.98	15.34	14.92	0.60	15.62	15.18
WM Top- T	0.00	12.60	18.65	0.00	11.81	20.66
Qwen3-VL-8B	11.58	17.49	25.99	15.08	21.28	31.00
Gemini 2.5 Pro	3.06	11.55	28.09	3.77	13.19	35.38
Ours	no hint	12.46	20.62	30.82	16.07	24.06
	w/ hint	9.40	21.38	31.55	10.32	21.53
					38.77	

Table 2: **Multi-step video procedure anticipation (VPA, %).** This is an open-loop evaluation without a candidate boundary. Seq-EM evaluates the complete action sequence, Pos.Acc evaluates actions by position, and Set IoU evaluates action composition while ignoring order. EGO’s gains arise mainly in composition and position prediction and exceed both repeated WM Top-1 and ranked Top- T outputs, so they cannot be explained by copying the world-model ranking.

Functional influence of task belief. We measure the functional influence of belief by replacing only the task belief while holding reasoning and observation fixed. *Belief-Induced Action Change* is the resulting action-change rate, while the *Belief-Swap Effect* subtracts the rate under a meaning-preserving paraphrase control. Belief-Induced Action Change rises to 27.1% for EGO but remains low for Base and Prospection, yielding a 24.3% Belief-Swap Effect (Figure 4(c)). Retrospection therefore makes task belief a functional input to selection rather than decorative text.

From grounded judgment to action selection. With correct-action exposure and training budget controlled, EGO exceeds No-Boundary by 6.6 pp in Within-Boundary Accuracy. Together, the results indicate that Prospection strengthens discrimination based on the predictive prior and action history, while Retrospection links task belief to selection and increases evidence use in generated traces.

From Reasoning to Planning

We evaluate whether judgment learned for single-step selection transfers to open-loop multi-step video procedure anticipation (Chang et al. 2020), where the model predicts the next T actions in order from one observation and action history (Table 2). The default EGO model receives no candidate boundary, isolating transfer beyond boundary-constrained selection.

Sequence Exact Match (Seq-EM) requires the entire ordered action sequence to be correct. Position-wise Accuracy (Pos.Acc) evaluates each predicted position, while Action-Set IoU (Set IoU) measures action composition without considering order. EGO improves mainly position and composition rather than exact-sequence recovery. It outperforms rules that repeat WM Top-1 or output the Top- T candidates by rank, excluding world-model rank copying as the explanation. A world-model hint for only the first action improves some partial metrics but not the overall plan consistently.

These results suggest that long-horizon planning requires predictions and judgments to be updated after execution

rather than extrapolated from a boundary constructed at one time point. EGO could support such closed-loop planning by updating the candidate boundary, task belief, and next action after each new observation. Whether iterative updates improve plan stability and success remains future work because the present evaluation is open-loop.

K	Cov.	World-Model	margin over the floor		E2E
	@ K	Top-1	Prospection	EGO	(%)
3	22.0	45.9	+4.9	+8.0	11.8
5	30.3	34.2	+4.0	+6.0	12.2
10	43.4	24.2	+4.6	+4.3	12.4

Table 3: **Boundary size.** Coverage@ K is the rate at which the annotated next action appears in the candidate boundary, and margin denotes improvement over WM Top-1 for each boundary. Increasing K raises both coverage and discrimination difficulty, leaving End-to-End Accuracy nearly unchanged. The video predictive model should therefore improve coverage within a compact, discriminable boundary rather than simply increasing the number of candidates.

Boundary Size

The candidate boundary constrains which actions EGO can select. Because changing K also changes the covered evaluation subset, we compare each subset’s margin over WM Top-1 together with End-to-End Accuracy (Table 3). Increasing K from 3 to 10 raises Coverage@ K from 22.0% to 43.4%, but the additional candidates increase discrimination difficulty, leaving End-to-End Accuracy nearly unchanged from 11.8% to 12.4%. EGO cannot recover the annotated action when it lies outside the boundary, yet simply enlarging the boundary does not resolve this limitation. The predictive model must instead place the annotated action more often within a compact, discriminable boundary.

Conclusion

Rather than fusing an independently trained video predictive model with a language model end to end, EGO uses its visually learned action prior as an explicit alignment signal. This alignment teaches the model to read action-sequence progression, discriminate among plausible candidates, and connect observable evidence and task belief to action selection. This selection behavior partially transfers from single-action prediction to subsequent-action composition, demonstrating the potential of modular embodied reasoning that separates and aligns experience-based prediction and language-based reasoning.

Measured performance is constrained by whether the predictive prior includes the annotated next action within a compact, discriminable boundary. Future work should learn more accurate and focused boundaries from diverse egocentric experience that captures human action patterns and cognitive biases. Extending EGO to closed-loop planning and continual learning could further enable embodied agents to update and expand their modeled world through sustained experience and adapt to changing environments.

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Appendix

This appendix consolidates the supplementary material accompanying this paper: implementation details, data contracts, evaluation protocols, and additional analyses needed to reproduce the reported EGO results. Executable code, configurations, prompt definitions, and metric implementations are available from the authors on request. The multi-step evaluation covers the open-loop video procedure anticipation setting reported in the main text.

S1 Why V-JEPA2 Action Anticipation

EGO requires a predictive model that maps the current observation and completed-action history to a ranked distribution over plausible next actions. This component must contribute evidence learned from visual experience independently of the VLM’s language prior, expose a discrete candidate interface, and avoid the cost and error of generating full future scenes. We instantiate this role with a frozen V-JEPA2 representation backbone and a separately trained action-anticipation head. The resulting Top- K predictions form the predictive action boundary $\mathcal{D}_t$ used for both VLM post-training and inference-time selection.

Why Representation-Based Action Anticipation Fits EGO

V-JEPA2 is particularly suitable for this interface because it predicts future representations rather than future pixels (Asran et al. 2025). Its large-scale video pretraining captures temporal and motion structure, while an action-anticipation head maps the frozen representation to a fixed action vocabulary. This gives EGO a ranked behavioral prior without requiring visual rollout generation, textual world simulation, or end-to-end fusion with the VLM. The prior should not be interpreted as a verifier of physical feasibility: it estimates behaviorally plausible actions under observed experience, while the VLM determines which candidate is consistent with the task context.

Strictly Future Next-Action Adaptation

The public V-JEPA2 EPIC-Kitchens-100 downstream protocol predicts the labeled segment’s own verb and noun from

EGO requirement	V-JEPA2 action-anticipation property
Visual predictive evidence	Future-oriented video representations learned independently of the VLM
No full-scene generation	Representation prediction avoids synthesizing task-irrelevant pixels
Discrete action interface	A supervised head produces a ranked distribution over GoalStep labels
Procedural context	A history module conditions predictions on completed visual segments
Modular alignment	The frozen predictor supplies candidates without end-to-end fusion

Table 4: Why V-JEPA2 action anticipation is a suitable predictive backbone for EGO.

a clip that extends to one second before that segment ends. The observed clip therefore overlaps substantially with the labeled action. EGO instead requires a strictly future prediction: if A_2 is the observed reference action, the target A_3 is the first subsequent action at the same annotation level and must satisfy

$$A_3^{\text{start}} \geq A_2^{\text{end}}.$$

No frame from A_3 is included in the observation. EGO also adds an ordered history of label-free visual features from completed same-level segments, allowing the predictor to represent procedural progress beyond the current clip. Table 5 summarizes the resulting change in prediction contract.

The V-JEPA2 encoder and predictor remain frozen. The attentive action head and history module are trained separately from every VLM alignment stage. Thus, “independently trained” denotes independence from the VLM rather than an entirely label-free predictive pipeline: self-supervised V-JEPA2 pretraining and supervised adaptation to the GoalStep action vocabulary are distinct stages. We use the action-free V-JEPA2 representation model, not the action-conditioned V-JEPA2-AC variant.

Predictive-Backbone Interface

The framework is defined by the interface

$$\begin{aligned} (x_{\leq t}, Z_{< t}) &\mapsto q_{\text{WM}}(a \mid x_{\leq t}, Z_{< t}) \\ &\mapsto \mathcal{D}_t = \text{TopK}_K[q_{\text{WM}}(\cdot \mid x_{\leq t}, Z_{< t})], \end{aligned} \quad (7)$$

not by a specific backbone architecture. Other video predictors could replace V-JEPA2 if they provide an independent ranked next-action distribution, but they would require task-specific adaptation of the action ontology, temporal granularity, history encoding, and score-to-candidate mapping. V-JEPA2 action anticipation is the current instantiation because it satisfies this interface directly while preserving the modular separation between experience-based prediction and language-based reasoning. Table 6 positions representative alternative model families at this interface level; it reports no accuracy figures and makes no empirical superiority claims.

S2 Reproducibility Overview

Pipeline and Artifact Flow

World-model grounding constructs a shuffled Top-10 predictive action boundary from the V-JEPA2-based next-action distribution. Prospection then aligns Qwen3-VL-8B to discriminate the annotated continuation within that boundary. Retrospection projects hindsight-confirmed task judgments into decision-time reasoning and task beliefs, while Prospection replay preserves candidate discrimination. The final adapter is evaluated both for single-step selection and, without additional training, for open-loop multi-step video procedure anticipation (VPA). The action vocabulary contains 81 verbs, 140 nouns, and 293 observed verb–noun labels from GoalStep.

Data Splits and Sample Flow

The original GoalStep training and validation videos are disjoint. Strict future-action indexing removes terminal rows

Aspect	Public V-JEPA2 EK100 setting	EGO instantiation
Prediction target	The observed segment’s own verb and noun	First strictly future same-level action A_3
Observed range	Ends 1.0 s before the labeled segment’s own end	Up to 8 s, ending 1.0 s before A_2 ends
Target-action frames	Included through temporal overlap with the labeled segment	Excluded by $A_3^{\text{start}} \geq A_2^{\text{end}}$
Completed history	Not used	Up to eight completed same-level segments, label-free visual features
Trainable module	Attentive probe over a frozen encoder and predictor	Attentive probe plus a history-context residual head over the same frozen backbone
Output to the VLM	Not applicable	Shuffled Top- K labels only; scores, ranks, and predictor identity hidden

Table 5: Public V-JEPA2 downstream prediction and the predictor used by EGO. The comparison describes the validated behavior of the public EPIC-Kitchens-100 configuration and the strictly future contract used in our experiments.

Model family and representative	Prediction target	Native output	Adaptation needed for EGO
Representation-based video predictor <i>V-JEPA2</i> (Assran et al. 2025)	Future latent representation	Features; action distribution via attentive probe	Current instantiation
Action recognition / anticipation <i>AVT on EPIC-Kitchens-100</i> (Girdhar and Grauman 2021; Damen et al. 2022)	Observed or near-term clip’s own verb and noun	Discrete action logits	Enforce a strictly future clip contract; add history conditioning
Action-conditioned latent world model <i>V-JEPA2-AC</i> (Assran et al. 2025)	Future latent state given a candidate action	State/energy prediction	Generate candidate actions; map scores to a discrete boundary
Generative video world model <i>Cosmos Predict</i> (Agarwal et al. 2025)	Future pixels/video; conditioning model-dependent	Visual rollout	Convert rollouts to semantic action candidates; higher interface complexity
Language-based world model <i>VLWM</i> (Chen et al. 2025)	Textual next action or state change; conditioning model-dependent	Language rollout	Recover an independent nonlinguistic prior and a calibrated candidate set

Table 6: Predictive model families and the EGO interface (interface-level comparison, not an empirical benchmark). Conditioning on a candidate action is stated inline in the prediction target where it applies.

Component	Training or generation unit	Output used downstream
World-model grounding	Strict future pair $A_2 \rightarrow A_3$	Deterministically shuffled Top-10 labels with scores and ranks hidden
Prospection	4,189 boundary-covered examples	Prospection adapter θ_{Pro}
Retrospection	2,347 accepted projected traces	Final single-step adapter sft_r15_c
VPA evaluation	Held-out sequences with $T \in \{3, 4\}$	One greedily generated JSON action sequence

Table 7: Final artifact flow using the terminology of the main text. VPA is an evaluation-only transfer setting.

without a subsequent same-level action. For VLM alignment and evaluation, the 128 videos in the predictor validation pool are partitioned with seed 42 into 38 development videos and a 90-video held-out evaluation cohort. The cohort is video-disjoint from all VLM development videos. The predictor checkpoint is selected using the fixed validation Action Top-5 criterion specified in the training recipe (Section S3); no VLM checkpoint or evaluation condition is selected on the 90-video cohort.

Artifact	Train/dev	Held-out	Notes
Candidate source segments	30,374	7,214	Segment bank
Strict future-action pairs	29,293	6,960	$A_2 \rightarrow A_3$
VLM held-out pool	—	5,326	90 videos
Boundary-covered examples	4,189*	2,313	*capped pool
Single-step evaluation	—	1,000	86 videos
VPA, $T = 3$	—	915	71 videos
VPA, $T = 4$	—	504	54 videos

Table 8: Sample flow. *The Prospection pool is formed by first sampling at most 30 examples per video, capped at 6,000, and then retaining the 4,189 examples whose annotated next action lies in the Top-10 boundary.

Checkpoint Provenance

Predictive action boundaries are generated by the history-conditioned action predictor described in Section S3. Single-step EGO results use `sft_r15_c`, the final Retrospection endpoint initialized from Prospection; the multi-step VPA transfer is evaluated with the Retrospection adapter `sft_r15` (Section S6). Full hashes and compute details are collected in Section S10.

S3 Predictive Action Boundary Construction

Prediction and Leakage Contract

Let A_2 denote the observed reference action and A_3 the first later action in the same video at the same annotation level. The target satisfies $A_3^{\text{start}} \geq A_2^{\text{end}}$. The visual observation is

$$[\max(0, A_2^{\text{end}} - 9\text{s}), A_2^{\text{end}} - 1\text{s}],$$

which spans at most eight seconds and excludes both the final second of A_2 and every frame of A_3 . The visual history $Z_{<t}$ contains up to eight recent same-level segments satisfying $A_h^{\text{end}} \leq A_2^{\text{start}}$. Ground-truth history labels are never provided to the predictor.

Strict future-action indexing removes 1,081 training and 254 validation terminal rows. The horizon from A_2 's end to A_3 's start has a median of 1.52 s and a 90th percentile of 20.00 s in training, and a median of 1.55 s and a 90th percentile of 25.69 s in validation. Action boundaries and annotation levels are treated as offline oracle annotations; deployment-time segmentation and hierarchy-estimation errors are outside the scope of this evaluation.

Frozen Visual Backbone and History Context

Each segment is represented by 32 uniformly sampled frames at 4 fps, resized to short side 256. The V-JEPA2 ViT-L encoder and predictor are frozen. A direct verb/noun/action head uses four attentive blocks with 16 heads and output sizes 81/140/293. The direct head was trained for eight epochs, and its epoch-3 checkpoint initializes the frozen visual prior used by the history-conditioned model.

For the current segment and up to eight completed segments, the stored feature tensor has shape $[B, 9, 17, 1024]$. A shared 16-head attentive pooler summarizes each segment. We add normalized recency $\log(1 + \Delta t) / \log(301)$, hierarchy-level, and slot embeddings, then process the sequence with a two-layer, 16-head Transformer with MLP ratio 4 and dropout 0.1. History-segment dropout is 0.3. The final logits are

$$z_f = z_f^{\text{visual}} + \tanh(g_f) z_f^{\text{context}}, \quad f \in \{\text{verb, noun, action}\},$$

where every gate g_f is initialized to zero.

Item	Final setting
Optimizer	AdamW, learning rate 3×10^{-4} , weight decay 10^{-4}
Schedule	One-epoch warmup, cosine decay to zero
Precision	bf16 training; fp32 validation
Batch / training length	Train 32, evaluation 64; 10 epochs for the history model
Loss	Fused verb/noun/action focal loss ($\gamma = 2$, $\alpha = .25$) plus 0.25 $\times$ history-only focal loss
Regularization	Segment dropout 0.3, gradient-norm clipping at 1.0
Selection	Validation Action Top-5, evaluated after every epoch

Table 9: Training recipe for the history-conditioned next-action predictor.

Boundary-Generating Checkpoint

All VLM boundaries are generated by the history-conditioned model at epoch 6. Its Action Top-1/5/10/15 accuracies are 10.45/30.66/43.92/52.64%. Coverage@10 is 43.92% on the full predictor validation set and 43.43% on the held-out evaluation cohort. This checkpoint was selected by the fixed per-epoch validation criterion of the training recipe (full-validation Action Top-5, Table 9); no ensemble or auxiliary transition objective is used. The complete checkpoint hash is reported in Table 26.

S4 Prospection and Retrospection

Common Input and Output Contract

All VLM conditions use Qwen/Qwen3-VL-8B-Instruct. Each example contains eight frames uniformly sampled from the same observation window used for world-model grounding, the 15 most recent completed actions in oldest-to-newest order, and a shuffled Top-10 predictive action boundary. Candidate order is deterministic from the sample identifier and fixed across splits. World-model probabilities and original ranks are retained only in the offline artifact and are never exposed to the VLM. The policy generates three fields: reasoning, task belief, and an action copied exactly from the candidate set.

No-Boundary uses the same frames, completed-action history, target actions, seed, and optimizer budget, but removes the candidate boundary and trains only the annotated action span. This controlled comparison isolates candidate-aligned learning from action-target exposure alone.

Prospection

From the 29,293 strict future-action training contexts, we first sample at most 30 examples per video and cap the pool at 6,000 using seed 42. The 4,189 examples whose annotated next action is contained in $\mathcal{D}_t$ form the Prospection set. For each candidate a , we compute the mean action-token log-likelihood under the same visual and history context,

$$s_\theta(a; c_t) = \frac{1}{|a|} \sum_{j=1}^{|a|} \log \pi_\theta(a_j | c_t, a_{<j}),$$

and minimize

$$\mathcal{L}_{\text{Pro}} = -\log \frac{\exp s_\theta(a_t^*; c_t)}{\sum_{\tilde{a} \in \mathcal{D}_t} \exp s_\theta(\tilde{a}; c_t)}.$$

Only the action span is scored; no reasoning prefix is teacher-forced. No-Boundary instead minimizes the negative mean log-likelihood of the annotated action span.

Retrospection: Offline Hindsight Projection

A parameter-frozen copy of the base Qwen3-VL-8B serves as the retrospective self. Reflection receives only the annotated sequence of at most five subsequent actions within 24 seconds and summarizes the local procedure that was underway at the decision point. Projection then receives this hindsight analysis, the decision-time frames, completed-action history, shuffled candidates, and annotated next action. It rewrites the

Item	Prospection / No-Boundary
Trainable parameters	LoRA $r = 16$, $\alpha = 32$, dropout 0.05, no bias; $q/k/v/o$ projections
Optimizer	AdamW, learning rate 10^{-5}
Schedule	One epoch, cosine decay, no warmup
Accumulation	8, gradient-norm clipping at 1.0
Precision	bf16 with gradient checkpointing
Seed / updates	42; 523 optimizer updates
Processed examples	4,188 processed, one skipped

Table 10: Prospection and No-Boundary training settings.

hindsight-confirmed judgment as reasoning and a task belief supported only by information available at decision time. The annotated action is appended afterward as the action target rather than generated by the projector.

The quality gate rejects the complete projected trace if parsing fails, reasoning contains fewer than 15 words, the belief contains fewer than three words, the belief restates or paraphrases the target action, or either field leaks future information or misstates temporal status. These per-trace rejection rules were fixed before trace filtering and applied unchanged to all 4,189 projected traces, retaining 2,347 traces (56.0%). The principal rejection categories are action restatement (1,064), future leakage in the belief (487), parse failure (448), future leakage in the reasoning (138), and temporal error (61).

Retrospection Optimization and Prospection Replay

Retrospection is initialized from θ_{Pro} . Accepted traces supervise reasoning, belief, and action with

$$\mathcal{L}_{\text{Retro}} = 0.5\mathcal{L}_{\text{reasoning}} + 1.0\mathcal{L}_{\text{belief}} + 0.25\mathcal{L}_{\text{action}},$$

where each field loss is averaged over the tokens in that field. Every projected-trace SFT micro-step is retained. With probability $\rho = 0.15$, it is followed by an additional Prospection replay micro-step drawn from the same boundary-covered pool. This schedule matches the additive replay definition in the main paper and prevents replay from replacing projected-trace exposure.

Item	Final Retrospection setting
Data	2,347 accepted traces, one epoch
Initialization / LoRA	θ_{Pro} ; same $r = 16$, $\alpha = 32$, dropout 0.05 modules
Replay	Additional Prospection step with Bernoulli probability $\rho = .15$
Optimizer	AdamW, learning rate 5×10^{-5}
Schedule	Cosine decay, no warmup; accumulation 8, clipping at 1.0
Precision / seed	bf16 with gradient checkpointing; seed 42
Training length	2,781 micro-steps and 347 optimizer updates

Table 11: Final Retrospection configuration for sft_r15_c.

Prompt Templates

Policy VLM system prompt (verbatim)

You are an embodied agent reasoning about your own ongoing activity from a first-person view. You see frames from the last 8 seconds of your first-person video, a list of actions you already COMPLETED, and a shuffled list of candidate next actions. Each action is “verb noun”. Exactly ONE candidate is what you do next (after the current action ends). Respond in EXACTLY this format: `<reasoning>` Compare the candidates against what you see and your completed-action history. 3–6 sentences. `</reasoning>` `<task_belief>` One sentence: the local procedure or subgoal you are currently in. Do NOT name the chosen next action verbatim. `</task_belief>` `<action>` verb noun `</action>`. The action line must copy one candidate EXACTLY as written.

The user message lists completed actions, shuffled candidates, and the next-action question in that order.

Hindsight reflection prompt (verbatim)

You are an activity-analysis assistant. You are given the sequence of actions a person performed AFTER a decision point in a cooking-style egocentric video. Infer what local procedure was underway AT the decision point. Return ONLY a JSON object with keys: {“activity”: str, “stage”: str, “completed_subgoal”: str, “next_subgoal”: str, “hypotheses”: [str, . . .], “uncertainty”: “low”|“medium”|“high”}. Keep each value short (≤ 12 words). “next_subgoal” is the immediate procedural step, NOT a verbatim action label.

Decision-time projection prompt (verbatim)

You write a decision-time rationale in the first person, as the camera wearer of an egocentric video. You see frames from the last 8 seconds BEFORE a decision point, the actions you completed so far, and a shuffled candidate list of next actions. You are also given (a) a hindsight analysis of the procedure inferred from what happened later, and (b) the action you actually did next. Write what you could have concluded AT the decision point, speaking as “I”, using ONLY evidence visible in the frames and your completed-action history.

1. Never mention or imply events after the decision point (no future actions, objects or outcomes that are not visible).
2. Never say the next action is already happening or done.
3. If the visible evidence is ambiguous, keep the belief at the level of the local procedural stage—do not overclaim the exact goal.
4. `<task_belief>` states the procedural stage you are IN, not what you will do next. It must NOT name the next action’s verb–noun label or restate that action in other words, in any grammatical form. Within that limit, make it specific enough to tell this moment apart from a neighbouring one: say which sub-goal is underway and how far it has got. Vary the wording naturally—do not reuse a fixed opening phrase.
5. `<reasoning>` must cite the concrete evidence you are judging from—including the actions you have already completed, whenever they bear on the choice—and state what that evidence implies about the current stage. Do not merely re-describe the scene, and do not turn the passage into a checklist that dismisses each candidate in turn.

Respond in EXACTLY this format: `<reasoning>` 3–6 sentences. `</reasoning>` `<task_belief>` One sentence about the current local procedure/subgoal. `</task_belief>`.

S5 Single-Step Evaluation Protocol

Evaluation Set, Decoding, and Parsing

From the 5,326 held-out contexts, 2,313 contain the annotated next action within the Top-10 boundary. We select a fixed 1,000-example subset using seed 42 and sample-identifier ordering. All conditions share the same 86 videos, examples, frames, completed-action histories, and candidate permutations. Generation is greedy with at most 320 new tokens.

The main results use a strict parser requiring closing tags for reasoning, task belief, and action. Action scoring first requires exact string match; normalized matching is allowed only when it identifies a unique candidate. All parse failures and invalid outputs count as incorrect. Intermediate adapters were stored and evaluated after training but were not selected by held-out performance. The reported EGO policy is the predefined 347-update Retrospection endpoint.

Metrics and Denominators

Metric	Definition
Coverage@ K	Fraction of examples whose annotated next action lies in the world-model Top- K boundary
Within-Boundary Accuracy	Correct selection among examples covered by the boundary
End-to-End Accuracy	Accuracy including boundary misses
WM-Correct Retention	Rate of preserving WM Top-1 when that prediction is correct
WM-Error Correction	Rate of recovering the annotated action when WM Top-1 is wrong but the action is covered
History-Removal Drop	Accuracy change after removing completed-action history
History-Replacement Drop	Accuracy change after replacing history with another video’s history
Evidence-Citation Gap	Accuracy difference between traces that cite completed actions and those that do not
Belief-Induced Action Change	Action-change rate after replacing only the task belief
Belief-Swap Effect	Belief-induced change minus a meaning-preserving paraphrase control

Table 12: Single-step metrics, using the terminology of the main text.

The Shared WM-Covered Set contains 1,000 examples from 86 videos and is common to every VLM condition. Coverage@10 on the full held-out pool is 43.43%. End-to-End Accuracy is computed as Coverage@10 multiplied by Within-Boundary Accuracy because VLM selection is evaluated on the covered subset. WM-Correct Retention is measured on the 242 examples for which WM Top-1 is correct; WM-Error Correction is measured on the remaining 758 covered examples.

Interventions and Uncertainty

For history removal, completed actions are deleted while frames and candidates remain fixed. For history replacement, the donor comes from another video and is matched by history length when possible, otherwise by the nearest available length, using a fixed seed. Candidate identities, order, and world-model scores remain unchanged.

For belief intervention, frames, reasoning, candidates, and decoding settings remain fixed while only the task belief is replaced. The Belief-Swap Effect subtracts the action-change rate produced by a meaning-preserving paraphrase from the rate produced by a belief taken from another example. All single-step confidence intervals use a paired cluster bootstrap over videos with $B = 2,000$. Because every model is trained with a single seed, these intervals do not include training-seed variance.

S6 Multi-Step Video Procedure Anticipation

Evaluation Construction

VPA is an evaluation-only transfer setting: no additional fine-tuning is performed. Starting from the 2,313 held-out contexts whose first annotated continuation lies in the Top-10 boundary, we retain examples with at least T future actions and no adjacent action gap larger than five seconds. This yields 915 examples from 71 videos for $T = 3$ and 504 examples from 54 videos for $T = 4$. Filtering uses only first-action coverage and does not guarantee that later positions are easier.

For target onset t , the visual input consists of eight frames uniformly sampled from $[t - 5, t - 1]$, including both endpoints. Images are resized to short side 336 only when larger. The model also receives the goal, the 15 most recent actions completed before the observation ends, and the full 293-label action vocabulary. It generates all T future actions in one open-loop pass without environmental rollout or intermediate visual updates.

VPA prompt contract (verbatim)

You are a procedural planning assistant for egocentric cooking videos. You are given: 8 frames sampled at 2 fps from the last 4 seconds of first-person video, ending 1 second before the next action begins; the goal; and the actions the camera-wearer has already COMPLETED. Predict the next T actions, in temporal order, starting with the action that begins immediately after the observation ends. Each action is “⟨verb⟩ ⟨noun⟩”. You MUST copy labels verbatim from the candidate list and output EXACTLY T of them as a JSON array of strings, and nothing else. The user message contains GOAL, ACTIONS ALREADY COMPLETED (in order), all CANDIDATE ACTION LABELS, and the same exact-length JSON instruction.

The text-only ablation replaces the visual-input clause with “no video (text context only)” while preserving the goal, completed actions, vocabulary, decoding, and all remaining prompt text. For EGO with a world-model hint, only the first predicted action is constrained to the shuffled Top-10 boundary; actions 2 through T remain unconstrained within the full vocabulary. Scores and original ranks are hidden. Figure 7 shows one stored card of the qualitative review

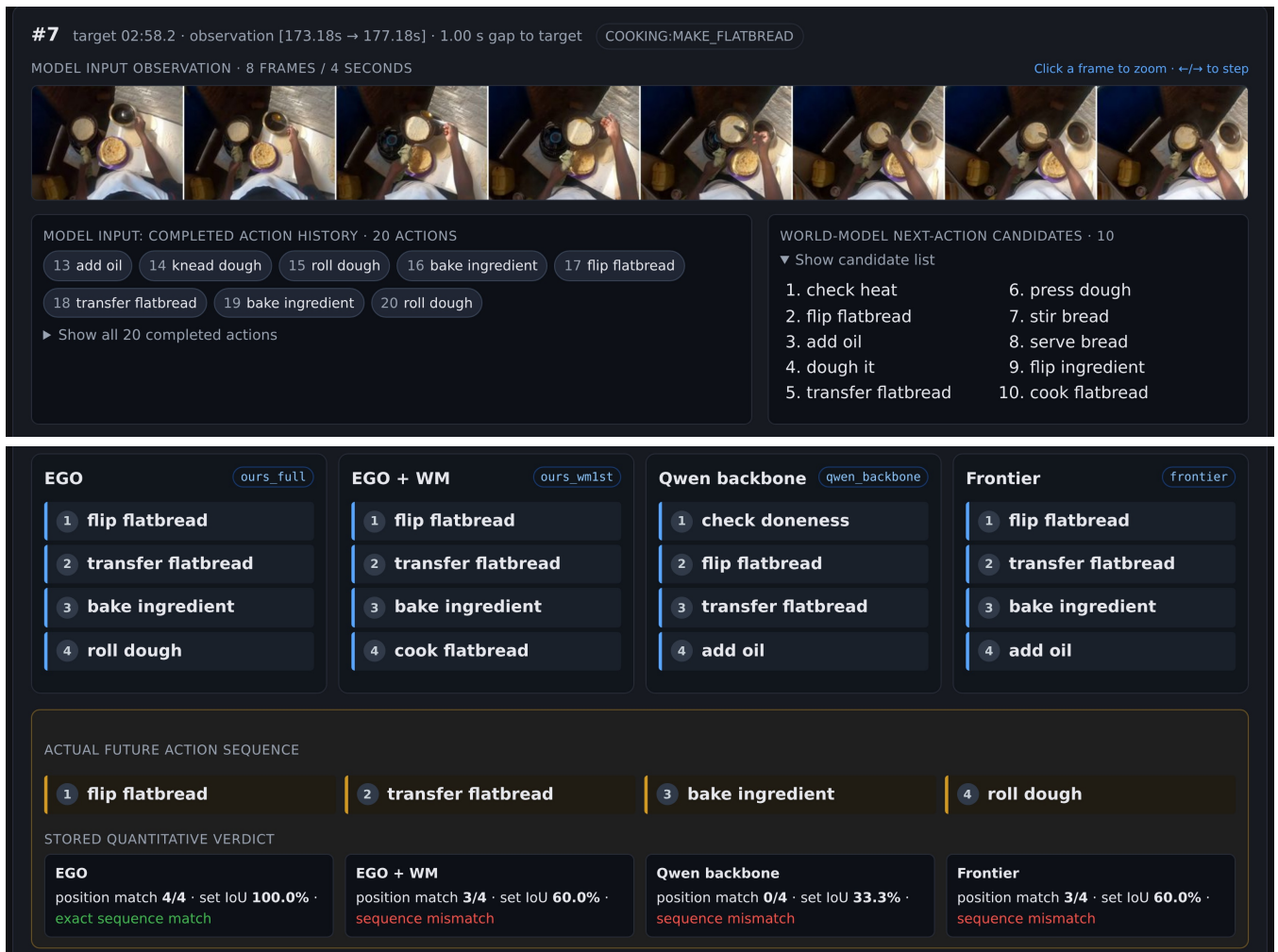


Figure 7: **VPA review card**. Two crops from one stored card of the $T = 4$ qualitative review site (COOKING:MAKE_FLATBREAD): the model input—eight frames spanning $[t - 5, t - 1]$, the completed-action history, and the shuffled world-model Top-10 candidates—and, below, the four arms’ raw parsed plans with the revealed ground truth and the stored per-arm verdict after the official vocabulary mapping. The interface labels the four arms EGO, EGO + WM, Qwen backbone, and Frontier, where Frontier is the Gemini 2.5 Pro arm of Table 13. In this example EGO matches all four positions, EGO + WM and Frontier match three, and the backbone matches none. Rating widgets and navigation elements are cropped out. The full artifact contains 504 examples from 54 videos and 2,016 model-plan cards.

interface with the exact inputs and the four arms’ plans for one example.

Checkpoint, Decoding, and Metrics

The VPA battery is evaluated with the Retrospection adapter `sft_r15`; the single-step battery uses the final Retrospection endpoint `sft_r15_c`. The two adapters follow the same Retrospection recipe, and their VPA metrics agree within 2 pp on every reported measure, well inside the video-cluster confidence intervals. Weight and configuration hashes for both appear in Table 26. No VPA-specific training is performed.

Local models use bf16 greedy decoding with temperature zero and at most 96 new tokens. After parsing the JSON array, labels are normalized with Unicode NFKC, lowercasing, and

whitespace/punctuation normalization. When necessary, a generated string is mapped to the nearest vocabulary item with cutoff 0.6. Missing positions and parse failures count as incorrect.

Sequence Exact Match (Seq-EM) requires the complete ordered sequence to match. Position-wise Accuracy (Pos. Acc.) evaluates each position, and Action-Set IoU (Set IoU) measures action composition while ignoring order. Confidence intervals use a video-cluster bootstrap with 1,000 replicates. Paired differences preserve the same examples and videos and use 2,000 paired video-bootstrap replicates with seed zero.

The $T = 3$ and $T = 4$ sets are different and should not be interpreted as a controlled horizon comparison. We do not claim consistent benefit from the world-model hint

Condition	Frames	Adapter	First action
WM Top-1 repeat	—	Rule	WM Top-1, repeated
WM ranked Top- T	—	Rule	WM ranks 1.. T
Qwen3-VL-8B	Yes	None	Full vocabulary
Qwen3-VL-8B, text only ($T = 3$)	No	None	Full vocabulary
EGO	Yes	sft_r15	Full vocabulary
EGO + world-model hint	Yes	Same	WM Top-10
Gemini 2.5 Pro	Yes	API model	Full vocabulary

Table 13: VPA conditions. The two WM rows are VLM-free rules over the world-model ranking (repeat Top-1 T times; emit ranks 1.. T). The two EGO conditions use identical weights and differ only in the first-action prompt constraint.

or superiority over Gemini. Because Seq-EM successes are concentrated in relatively few videos, Pos. Acc. and Set IoU provide important complementary evidence.

S7 Additional Results

Single-Step Selection

	WB-Acc	Paired Δ (95% CI)		WM-Corr.	WM-Err.
	(%)	vs. No-Boundary	vs. WM Top-1	Reten.	Correc.
Base VLM	21.0	—	—	29.3	18.3
No-Boundary	21.9	—	—	26.0	20.6
Prospection	28.8	+6.90 [+3.22, +10.84]	+4.60 [−0.35, +9.74]	35.5	26.7
EGO	28.5	+6.60 [+3.48, +9.48]	+4.30 [−1.07, +9.49]	41.7	24.3
WM Top-1 floor	24.2	—	—	—	0

Table 14: Shared WM-Covered Set ($n = 1,000$, 86 videos). Malformed outputs count as incorrect. The controlled comparison is against No-Boundary, which receives the same targets and training budget without the predictive action boundary. Both intervals relative to WM Top-1 include zero.

Malformed-output rates are 1.7% for Base VLM, 1.2% for No-Boundary, 4.3% for Prospection, and 2.8% for EGO.

Training Dynamics

Prospection update	100	200	300	400	500	Final (523)
Prospection	22.4	24.4	26.4	27.7	27.6	28.8
No-Boundary	20.8	21.2	20.9	21.0	21.4	21.9

Table 15: Within-Boundary Accuracy (%) over Prospection training.

Intervention Values and Boundary-Size Sweep

Tables 17–19 record the point values plotted in main-text Figures 5 and 4(c) and Table 3; the intervention protocols are fixed in Section S5.

Retro update	Cumulative	WB-Acc	Retention	Correction	Malformed	Δ vs. final, 95% CI
100	623	29.0	39.3	25.7	5.0	+0.5 [−3.3, +4.7]
150	673	28.1	35.5	25.7	4.7	−0.4 [−4.3, +3.7]
200	723	29.2	44.2	24.4	3.2	+0.7 [−2.2, +3.7]
250	773	27.2	38.0	23.7	2.8	−1.3 [−4.1, +1.7]
300	823	27.1	36.8	24.0	2.6	−1.4 [−3.8, +1.1]
347	870	28.5	41.7	24.3	2.8	Reported endpoint

Table 16: Retrospection checkpoints with $\rho = .15$. Every paired interval relative to the final endpoint includes zero; the reported model is the predefined endpoint rather than the best held-out checkpoint.

	Kept	Removed	Replaced	Δ replaced
Base VLM	21.0	18.1	19.9	−1.1
No-Boundary	21.9	17.4	19.7	−2.2
Prospection	28.8	18.7	18.8	−10.0
EGO	28.5	18.7	17.9	−10.6

Table 17: History interventions on the Shared WM-Covered Set (WB-Acc, %; main-text Figure 5). “Kept” uses the original completed-action history, “Removed” deletes it (History-Removal Drop), and “Replaced” substitutes donor-video history; Δ is Replaced minus Kept (History-Replacement Drop).

	Belief swap	Paraphrase control	Net effect
Base VLM	3.2	1.6	1.6
Prospection	6.1	0.4	5.7
EGO	27.1	2.8	24.3

Table 18: Belief interventions (action-change rate, %; $n = 247$; main-text Figure 4(c)). The swap replaces only the task belief, the control applies a meaning-preserving paraphrase, and the net effect (swap minus control) is the Belief-Swap Effect.

K	Coverage@ K	WM Top-1	Prospection	EGO	E2E (EGO)
3	22.0	45.9	50.8	53.9	11.8
5	30.3	34.2	38.2	40.2	12.2
10	43.4	24.2	28.8	28.5	12.4

Table 19: Boundary-size sweep (%; absolute WB-Acc values behind main-text Table 3). Each K row is evaluated on its own covered subset, so WB-Acc values are not comparable across rows; E2E multiplies EGO’s WB-Acc by Coverage@ K .

Candidate-Free Generation Diagnostic

We remove the candidate list only at evaluation time and freely generate one action span on a fixed 500-example subset of the Shared WM-Covered Set. GT-Match requires an exact normalized verb–noun match; no candidate-based token-overlap fallback is used. In-Support measures whether the freely generated action lies in the hidden Top-10 boundary.

	In-Support (strict, %)	GT-Match (strict, %)	Malformed (%)
Base VLM	29.2	11.8	4.0
No-Boundary	44.0	15.2	0.4
Prospection	38.4	18.6	2.4
EGO	28.0	12.8	17.8

Table 20: Candidate-free generation, $n = 500$ per condition. Strict normalization is used for both In-Support and GT-Match.

No-Boundary most often generates an action within the hidden world-model support, yet it performs worst when explicitly discriminating among visible candidates in Table 14. This separates broad distribution matching from within-boundary discrimination. EGO does not improve unconstrained generation and has the highest malformed rate; the Retrospection gain is instead reflected in belief-conditioned selection and evidence use.

Prospection Replay Ablation

Table 21 varies the replay probability ρ . Removing replay changes selection by -2.3 pp, 95% CI $[-5.0, +0.7]$, and WM-Correct Retention by -7.0 pp, $[-15.2, 0.0]$. Because both intervals include or touch zero, we do not claim statistical significance.

ρ	Coverage@10	WB-Acc	Retention	Correction	E2E
0 (no replay)	43.4	26.2	34.7	23.5	11.4
0.15 (default)	43.4	28.5	41.7	24.3	12.4
0.30	43.4	27.6	40.5	23.5	12.0

Table 21: Prospection replay probability (%).

VPA Aggregate and Paired Results

For $T = 3$, EGO improves over Qwen3-VL-8B by $+0.87$ pp Seq-EM, $+3.13$ pp Pos. Acc., and $+4.84$ pp Set IoU. Removing video from Qwen3-VL-8B decreases Pos. Acc. and Set IoU by 8.31 pp and 12.23 pp, respectively. For $T = 4$, EGO improves Pos. Acc. by 2.78 pp and Set IoU by 6.48 pp, while the Seq-EM difference is not significant. Removing completed-action history reduces Pos. Acc./Set IoU by $8.04/14.26$ pp for EGO and $10.66/15.41$ pp for EGO with the world-model hint.

S8 Trace Diagnostics

These diagnostics are descriptive rather than causal. The rule-based detectors were not validated against human annotations; their precision and recall are therefore unknown.

History Evidence and Visual Cues

The history-evidence detector counts explicit references to completed or earlier actions, repeated action patterns, or trajectory progression. Generic scene description without a temporal reference does not count. The Evidence-Citation Gap compares accuracy between parseable traces that cite such evidence and those that do not.

	Citation rate	Acc. with	Acc. without	Gap (pp), 95% CI
Base VLM	22.4	31.8	18.3	$+13.5$ [$+4.0, +20.4$]
No-Boundary	37.8	24.1	21.0	$+3.2$ [$-5.5, +10.6$]
Prospection	32.4	38.4	26.1	$+12.3$ [$+4.2, +18.9$]
EGO	13.4	39.2	27.8	$+11.4$ [$+2.8, +21.4$]

Table 23: Evidence-citation rate and within-condition accuracy association (%). Only parseable reasoning traces are included, giving $n = 983/988/957/972$ for Base, No-Boundary, Prospection, and EGO, respectively.

Visual-Cue Mention is computed separately for two binary categories. A *spatial cue* locates the camera wearer or an object relative to another object or region, such as “on the stove” or “near the pot.” A *hand/tool cue* explicitly describes a hand, a held object, or tool use, such as “holding the ladle.” These definitions match the annotations illustrated in Table 1 of the main text.

Condition	Spatial cue (%)	Hand/tool cue (%)
Base VLM	34.2	33.8
No-Boundary	21.7	9.9
Prospection	27.5	19.5
EGO	71.0	65.2

Table 24: Visual-cue mention rates used in the main-paper analysis.

Rejected Surface-Form Controls

We also tested causal connectives, first-person wording, and contrastive connectives as surface-form controls. None serves as a reliable capability indicator: first-person use is saturated by the EGO prompt, contrastive connectives are too rare for stable comparison, and causal connectives do not positively distinguish correct from incorrect predictions.

Rate (%)	Base	No-Boundary	Prospection	EGO
Causal connective	16.2	5.1	10.4	27.4
First person	92.9	30.6	73.0	100.0
Contrastive connective	0.2	0.2	0.3	0.0

Table 25: Surface-form controls not adopted as capability metrics.

For causal connectives, the within-condition accuracy gap is -12.7 pp for No-Boundary, 95% CI $[-21.7, -3.0]$, and -5.1 pp for EGO, $[-13.8, +3.7]$. In Prospection, the first-person gap is -11.3 pp, $[-19.2, -3.3]$; no within-condition contrast is available for EGO because the rate is 100%. Contrastive-connective rates are at most 0.3% in every condition.

S9 Additional Reasoning Traces

Table 1 of the main text presents one example selected by a fixed criterion defined before inspection: the annotated action is inside the boundary; WM Top-1, Base VLM, and No-Boundary are wrong; Prospection and EGO are correct; the target is not a verbatim repeat of the immediately preceding

Method	$T = 3$: 915 examples, 71 videos			$T = 4$: 504 examples, 54 videos		
	Seq-EM	Pos. Acc.	Set IoU	Seq-EM	Pos. Acc.	Set IoU
WM Top-1 repeat	0.98	15.34	14.92	0.60	15.62	15.18
WM ranked Top- T	0.00	12.60	18.65	0.00	11.81	20.66
Qwen3-VL-8B	11.58	17.49	25.99	15.08	21.28	31.00
Gemini 2.5 Pro	3.06	11.55	28.09	3.77	13.19	35.38
EGO	12.46	20.62	30.82	16.07	24.06	37.48
EGO + world-model hint	9.40	21.38	31.55	10.32	21.53	38.77

Table 22: Multi-step video procedure anticipation (%). Seq-EM is exact ordered sequence match, Pos. Acc. is position-wise accuracy, and Set IoU is action-set overlap. Bold indicates the highest point estimate within a column and does not imply statistical significance. Each horizon uses a different subset satisfying first-action coverage and future-length requirements.

Artifact	Identifier	SHA256
Boundary predictor	history_k8_epoch6	1d20c942e7ec71ef76326679c44d20dead88e084dc8ac4367b7e3ebebdaa61cd
Prospection adapter	θ_{Pro}	dd73d0733d0950522419aa5df46f0756383f9cec5c049a50d2d479de19436beb
Single-step EGO	sft_r15_c	6da2c77458201771841f52cf72b35f27b11125ce0376b3d2aeb8847e0b6a6e73
VPA adapter	sft_r15	d33d695d6432e61a849320dce9f98d35c5c615aa9dbd07ae62d428bcc0e1cc9d
VPA adapter config	sft_r15_config	3249fb68a3cc9afa655b260196401466df242c2d38f1e31f83edc0e6e1de3734

Table 26: Checkpoint and configuration identities for the reported results.

action; and every displayed trace is well formed. We show three further verbatim examples satisfying the same criterion, drawn from distinct videos, followed by one failure mode in which EGO selects the correct action but its generated reasoning is not aligned with that action. These examples are illustrative; aggregate and intervention results provide the quantitative evidence.

Success A — GT roll dough (WM Top-1: *add topping*)

Candidates: add topping, roll dough, add ingredient, make dough, knead dough, check heat, sprinkle flour, put dough, cut dough, shape dough. Recent history: flip ingredient → add ingredient → roll dough → knead dough → roll dough → flip ingredient

Base VLM → *shape dough* ×

belief: Shape the flattened dough into the desired form.

reasoning: I have been repeatedly rolling and flipping the dough, and just added an ingredient. The dough is now spread out on the table, and I am currently pressing down on it with my hands, likely to flatten it further or prepare it for the next step. Given the context of preparing dough for a dish, the most logical next step is to shape it, as rolling has already been done multiple times and the dough is ready for forming.

Prospection → *roll dough* ✓

belief: Roll the dough to prepare it for the next step.

reasoning: I have been repeatedly rolling and flipping the dough, and adding ingredients. The current state shows the dough is on the table, and I’ve just flipped the ingredient. The most logical next step is to continue rolling the dough to achieve the desired thickness or texture.

EGO → *roll dough* ✓

belief: The dough has been kneaded and is now being prepared for the next stage of shaping or cooking.

reasoning: I have just finished kneading the dough and it’s now resting on the table, visibly flattened and spread out. The rolling pin is nearby but unused, and the ingredients are still in the bowl, suggesting the dough is ready for the next stage. Since I’ve been repeatedly rolling and flipping the dough, and the current state shows it’s been worked enough, the logical next step is to prepare it for shaping or cooking.

Success B — GT cook flatbread (WM Top-1: *transfer flatbread*)

Candidates: cut dough, flip flatbread, cook flatbread, transfer flatbread, sprinkle flour, press dough, roll dough, check heat, add oil, shape dough. Recent history: flip ingredient → roll dough → check heat → flip ingredient → roll dough → check heat

Base VLM → *flip flatbread* ×

belief: Flipping the flatbread to cook the other side.

reasoning: The current frame shows the flatbread being flipped on the hot surface, indicating the cooking process is ongoing. The previous action was ‘cook flatbread’, and the sequence suggests a repetitive cycle of cooking, flipping, and preparing the next batch. The most logical next step is to continue the cooking process by flipping the flatbread again to ensure even cooking on both sides.

Prospection → *cook flatbread* ✓

belief: Cooking the flatbread on the griddle.

reasoning: The current action history shows a repetitive cycle of rolling dough, checking heat, and cooking flatbread. The most recent action was checking heat, and the flatbread is still on the griddle. The logical next step is to continue cooking it, which aligns with the established pattern of the task.

EGO → *cook flatbread* ✓

belief: I am in the process of cooking a flatbread on a hot surface.
reasoning: I’ve just finished pressing the dough and am now holding the shaped flatbread, ready to place it on the hot surface. The fire is active and the cooking surface is visibly hot, indicating it’s the right time to begin cooking. I’ve been repeatedly checking the heat and flipping the ingredient, which suggests I’m in a cycle of preparing and cooking flatbreads. The dough is ready and the surface is hot, so the next step is to start cooking it.

Success C — GT add ingredient (WM Top-1: *add pepper*)

Candidates: cut ingredient, add oil, add pepper, check heat, wash ingredient, dispose item, cut pepper, add vegetable, add ingredient, unpack ingredient. Recent history: bake ingredient → cut ingredient → cut bean → add topping → spread ingredient → spread ingredient

Base VLM → *add vegetable* ×

belief: Adding lettuce to the sandwiches.

reasoning: I am currently adding lettuce to sandwiches, which indicates I am in the process of assembling them. The bag of lettuce is open, and I have already added some leaves. The most logical next step is to continue adding more lettuce to the sandwiches to ensure they are well-filled.

Prospection → *add ingredient* ✓

belief: Adding more ingredients to the sandwich.

reasoning: I have been assembling a sandwich by adding various ingredients. The current frame shows me placing lettuce on the sandwich, indicating I am still in the process of assembling it. The most logical next step is to continue adding ingredients to complete the sandwich.

EGO → *add ingredient* ✓

belief: I am in the process of assembling the sandwich and am considering the next ingredient to add.

reasoning: I have just finished adding a leafy green from the bag to the sandwich, and the bag is still open on the counter. My hands are now free, and I’m looking at the ingredients around me, including the jar of mustard and the container of sliced cheese. The sandwich is partially assembled, and I’m likely considering what else to add next, possibly from the items already out.

Failure mode — correct action with inconsistent reasoning (GT add rice)

Candidates: check heat, add rice, add spice, stir ingredient, add ingredient, turn stovetop, flip ingredient, mix ingredient, stir dough, add oil. Recent history: . . . add rice → add rice → add rice → add rice → add rice → check heat

EGO → *add rice* ✓

belief: I have just removed the cooked item from the pan and am now preparing to clean the ladle or set up for the next step.

reasoning: I have just finished transferring the cooked item from the pan to a metal bowl, and the pan is now empty except for some residue. The batter in the pan is still liquid and uncooked, indicating the cooking cycle for this batch is complete. I am holding the ladle and the bowl, positioned to either clean the ladle or prepare for the next step.

The selected action is correct, but the belief and reasoning do not support adding rice. This case shows why correct selection alone is insufficient evidence of reasoning–action alignment and motivates reporting aggregate interventions alongside qualitative traces.

Item	Final environment or measured runtime
GPU	One NVIDIA H200 143GB per job
Host memory	Container cgroup limit of 240 GiB
Software	Python 3.11; PyTorch 2.6.0+cu124; Transformers 5.9.0; PEFT 0.19.1; TRL 1.5.1
Distributed stack	None; single-GPU PyTorch loops
Predictive features	30,374 clips in 1 h 47 min
History-conditioned predictor	10 epochs in approximately 15.5 min after feature extraction
Prospection	3.65 h
Hindsight projection	0.74 h
Retrospection	0.69 h
Single-step evaluation	Approximately 556 s for the final 1,000-example battery

Table 27: Measured compute for the experiments reported in the paper.

S10 Compute and Artifact Manifest

The public code release accompanying this paper contains the source for world-model grounding, Prospection, Retro-spection, and VPA evaluation; final configurations; generic launchers; prompt definitions; metrics; schemas; checksum manifests; and an execution-order README. Licensed GoalStep videos and model weights are not redistributed and must be obtained under their original terms. Checkpoints, raw outputs, and run logs are excluded from the release; the hashes above identify the exact artifacts used for the reported results.